

Name: _____ Date: _____ Period: _____

1. For each rational function, state which form (**factored** or **standard**) is more useful for the requested task, then answer the question.

(a) $r(x) = \frac{x^2 + 3x - 10}{x^2 - 4}$. Find all zeros, holes, and VAs.

Best form: _____

Factor numerator: _____ Factor denominator: _____

Zeros: _____ Holes: _____ VAs: _____

(b) $s(x) = \frac{3x^5 - 2x^2 + 1}{4x^3 - x}$. Describe the end behavior.

Best form: _____

As $x \rightarrow +\infty$: $s(x) \rightarrow$ _____ As $x \rightarrow -\infty$: $s(x) \rightarrow$ _____

2. Perform polynomial long division. Show all steps.

Divide $f(x) = 3x^3 - 5x^2 + 2x - 4$ by $g(x) = x - 2$.

Work space:

$$q(x) = \underline{\hspace{2cm}} \quad r = \underline{\hspace{2cm}}$$

Write as $f(x) = g(x) \cdot q(x) + r$: _____

3. A rational function is defined by $R(x) = \frac{x^2 + x - 6}{x - 1}$.

- (a) Perform polynomial long division.

$$R(x) = \underline{\hspace{2cm}}$$

- (b) State the equation of the slant asymptote. Explain in one sentence why this is the slant asymptote.

Slant asymptote: $y =$ _____

- (c) What is the remainder when $x^2 + x - 6$ is divided by $x - 1$? What does this tell you about $R(1)$?

Remainder: _____ $R(1) =$ _____

4. Use Pascal's Triangle to expand each expression. Show the row of Pascal's Triangle you used.

(a) $(x + 3)^3$

Row used: _____

$$(x + 3)^3 = \underline{\hspace{2cm}}$$

(b) $(2x - 1)^4$ (Hint: let $a = 2x$ and $b = -1$)

Row used: _____

$(2x - 1)^4 =$ _____

5. **(AP-Style Multiple Choice)** Which of the following statements about $p(x) = \frac{x^3 - 8}{x - 2}$ is correct?

- (A) p has a vertical asymptote at $x = 2$.
- (B) p has a hole at $x = 2$ with y -coordinate 12.
- (C) p simplifies to $x^2 - 2x + 4$ for all real x .
- (D) p has a slant asymptote of $y = x^2 + 2x + 4$.

Answer: _____ Show your reasoning:
